CSE 202 Midterm 1 Part (a)

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I. Question (Algorithm Analysis) (15 points)

Given the following function

```
int algorithm1(int N, int k){
    if (N == 0)
        return 0;
    int sum = 0;
    for (int i = 0; i < N; i++)
        sum++;
    for (int i = 0; i < N; i++)
        for (int j = i; j < N; j++)
            sum++;
    for (int i = 0; i < 4; i++)
        sum += algorithm(N / 2);
    return sum;
}
```

What is the time complexity of algorithm1? Explain your result. Use the master theorem.

II. Question (Algorithm Analysis) (15 points)

Given the following function

```
int algorithm2(int N){
    int sum = 0;
    for (int i = 1; i <= N; i ++){
        for (int j = i; j <= N; j++){
            sum++;
        }
    }
}
```

What is the value of sum after the execution of the algorithm2?

III. Question (LinkedList) (15 points)

Write the method

```
void addAfterEachNode(Node<T> newNode)
```

which add newnode after each node in the singly linked list.

IV. Question (Stack) (15 points)

Implement the stack as a doubly linked list. Your implementation must only contain a head node of type DoubleNode, and three functions `boolean isEmpty()`, `DoubleNode<T> pop()` and `void push(DoubleNode<T> newNode)`.